

Firewall — Your Network's Security Guard

Study notes

1. What Is a Firewall?

A firewall is a security system that sits between your network and the outside world (usually the internet) and decides what traffic gets in and what stays out. Think of it as a security guard at a gate: every visitor — every data packet — gets checked against a set of rules before being allowed through. Match an "allow" rule, and it gets in. Match a "block" rule, or match nothing at all, and it's stopped.

2. How a Firewall Works

Traffic arrives → the firewall checks it against its rule list → the traffic is **Allowed** or **Blocked** → the decision is logged. This happens for every single packet trying to enter or leave the network.

3. How Firewall Rules Are Processed

Rules are checked in order, top to bottom. The moment a packet matches a rule, that rule's action is applied and checking stops — this is called **first match wins**. That's why rule order matters: a broad rule placed too early can accidentally catch traffic a more specific rule further down was meant to handle differently.

Source IP	Destination IP	Port	Protocol	Action
Any	192.168.1.0/24	80	TCP	Allow
Any	Any	23	TCP	Block

4. Types of Firewalls

Type	OSI Layer	What It Does
Packet Filtering	Layer 3–4	Checks IP, port, and protocol on each packet individually. Fast, but has no memory of past packets.
Stateful Inspection	Layer 3–4	Tracks active connections in a state table; only allows traffic belonging to an approved session.
Proxy (Application)	Layer 7	Fully inspects the actual content of a request before passing it on; acts as a middleman between client and server.
Next-Gen (NGFW)	Layer 3–4 & 7	Combines stateful filtering with application awareness, intrusion prevention, and deeper inspection.

5. What's a DMZ?

A DMZ (demilitarized zone) is a separate network segment for anything that must be reachable from the internet — like web or mail servers — kept apart from the actual internal network. If a DMZ server is compromised, the attacker still doesn't get a direct path inside, because a second firewall sits between the DMZ and the internal network.

Quick link: firewalls often sit at the same point in the network as NAT/PAT (Episode 10) — but they do different jobs. NAT translates addresses so devices can share one public IP. A firewall decides which traffic is allowed to use that connection at all.

6. Advantages & Limitations

Advantages

- Blocks unauthorized access
- Controls what traffic is allowed
- Logs activity for investigation

Limitations

- Can't stop threats starting inside the network
- Can't inspect encrypted traffic without extra tools
- Misconfigured rules can block good traffic or miss bad traffic

Separates trusted and untrusted networks

Advanced firewalls (NGFW) add cost and can slow traffic

7. Firewall Logs

Every allow/block decision gets recorded as a log entry — typically the source IP, destination IP, port, protocol, action taken, and timestamp. These logs are the first thing anyone investigating suspicious activity looks at.

8. Quick Interview Q&A

Q: What's the main purpose of a firewall?

A: To monitor and control incoming and outgoing network traffic based on a defined set of security rules.

Q: What's the difference between a packet filtering and a stateful firewall?

A: Packet filtering checks each packet on its own, with no memory of previous packets. Stateful tracks the state of active connections and only allows traffic belonging to an approved session.

Q: Why does the order of firewall rules matter?

A: Most firewalls process rules top to bottom and stop at the first match. A rule placed too early can accidentally catch traffic a more specific rule further down was meant to handle differently.

Q: What is a DMZ, and why do organizations use one?

A: A network segment for internet-facing servers, kept separate from the internal network, so a compromised DMZ server doesn't give an attacker a direct path inside.

9. Memory Hook

Firewall = your network's security guard. Every packet gets checked. First matching rule wins. Every decision gets logged.

10. Try It Yourself

● **Windows:** Start → search "Windows Defender Firewall"

● **Mac:** System Settings → Network → Firewall

Check what's currently allowed through, and what's blocked.